BEE 4750/5750

Mini-Project 2: Dispatching New York's Grid Under a Carbon Price

2026-10-31

Due Date
Friday, 11/20/26, 9:00pm

Overview

In 2013, New York's grid still burned coal, ran the Indian Point nuclear plant, and had almost no solar. In this project you will model how that grid met demand hour by hour, find the price of electricity each hour, and ask what impact a carbon price would have made: first by changing which plants run, and then by changing what gets built.

Instructions

- Students in BEE 4750 may work in groups of 2; students in BEE 5750 work individually.
- Submit a **report of no more than 4 pages**, not counting figures, tables, code, and references, as a PDF to Gradescope. Answer every numbered question in the report and label each answer with its number (for example, **3.2**), then tag those pages in Gradescope.
- Put your code in an appendix or submit the notebook alongside the report. The report should read on its own without it.
- Parts 1–4 use linear programming, shadow prices, JuMP, and multi-period dispatch (October 14–28 and the October 26 lab). Part 5 uses capacity expansion (November 2).

Load Environment

The following code loads the environment and makes sure all needed packages are installed. This should be at the start of most Julia scripts.

```
import Pkg
Pkg.activate(@__DIR__)
Pkg.instantiate()
```

```
using JuMP
using HiGHS
```

```
using CSV
using DataFrames
using Statistics
using Plots
using Measures
```

The System

We treat New York as a single node (no transmission limits) and model every hour of 2013.

Demand is NYISO's hourly load for 2013, summed over its eleven zones. It ranges from about 12 GW overnight to a 34 GW summer peak.

The existing fleet is New York's summer 2013 generating capability, grouped by type. Each plant type can generate up to its available capacity in each hour. The thermal types (nuclear, coal, steam, combined cycle, and combustion turbines) can also change their output by at most their ramp fraction of capacity from one hour to the next. Hydro, wind, and solar have no ramp limit.

Table 1: New York's 2013 generating fleet.

Type	Capacity (MW)	Variable cost ($/MWh)	CO_2 (t/MWh)	Ramp (per hour)	Availability
Nuclear	5,411	8	0	0.25	88%
Hydro	4,276	0	0	–	budget
Coal	1,548	24	1.00	0.70	100%
Oil and gas steam	9,576	45	0.58	0.70	100%
Combined cycle	8,991	30	0.43	0.70	100%
Combustion turbine	4,911	40	0.55	1	100%
Wind	1,634	0	0	–	profile
Solar	31.5	0	0	–	profile

The fleet file has a ninth row, combined cycle with carbon capture, with no capacity: New York had none in 2013. It generates nothing in Parts 1–4 and becomes an option in Part 5.

Hydro can be dispatched in any hour up to its capacity, but its total generation over the year cannot exceed **24,572 GWh**, New York's hydro generation in 2012. Most of that capacity is the New York Power Authority's plants on the Niagara and St. Lawrence rivers. At Niagara, a 1950 treaty with Canada sets how much of the river's flow must go over the falls rather than through the turbines, which limits the water available for power. Water used in one hour is not available in another.

Wind and solar follow hourly capacity-factor profiles, rescaled so their annual averages match New York's observed values: 23.4% for wind and 16.5% for solar.

Non-served energy is allowed but penalized at $10,000/MWh, so the problem is always feasible. A good plan should have none.

> **i** Data and Simplifications
>
> Capacities are from NYISO's *2013 Load & Capacity Data* report (the "Gold Book"), Table III-3a, summer capability; wind and solar use nameplate capacity. Coal, combined cycle, and combustion turbine variable costs are roughly what these plants cost to run at 2012–13 fuel prices, and their emissions are the values used in lecture; the others are course assumptions. The wind capacity factor is from the Gold Book's 2012 energy for wind plants in service all year, and the solar capacity factor is the Long Island Solar Farm's in NYISO's 2024 Gold Book. The ramp fractions are hourly limits: coal's and combined cycle's are the rates of the large units in the RTS-GMLC test system (about 4 MW per minute on a 350 MW unit), steam is assumed to match coal, combustion turbines can reach full output within an hour, and nuclear's is a course assumption.
>
> The model leaves out imports (New York imported roughly 15% of its electricity), pumped storage (1,407 MW), about 400 MW of small biomass and waste plants, and transmission limits between zones.

For comparison, in 2012 New York generated 139,137 GWh in-state: 29.4% nuclear, 17.7% hydro, 2.2% wind, and 48.1% fossil, of which coal was 3.1%.

The data, ready to use:

```
# Statewide hourly demand: the sum of NYISO's eleven load zones [MW]
load_data = DataFrame(CSV.File("data/2013_hourly_load_NY.csv"))
zones = ["A", "B", "C", "D", "E", "F", "G", "H", "I", "J", "K"]
demand = zeros(nrow(load_data))
for zone in zones
    demand .+= load_data[:, zone]
end

# Hourly wind and solar capacity factors, rescaled to New York's annual averages
profiles = DataFrame(CSV.File("data/wind_solar_capacity_factors.csv"))
wind_cf = profiles.Wind .* (0.234 / mean(profiles.Wind))
solar_cf = profiles.Solar .* (0.165 / mean(profiles.Solar))

# The existing fleet, as in the table above. Wind and solar availability come
# from the hourly profiles, not the Availability column. RampFraction is missing
# for hydro, wind and solar, which have no ramp limit. The last row, combined
```

```
# cycle with carbon capture, has no capacity until Part 5.
fleet = DataFrame(CSV.File("data/nyca_2013_fleet.csv"))

hydro_budget = 24_572_000   # annual hydro energy [MWh]
nse_cost = 10_000           # penalty on non-served energy [$/MWh]
```

Problems (Total: 100 Points)

Part 1: Formulate the Dispatch Problem (15 points)

Problem 1.1 (10 points)

Formulate a linear program that decides how much each plant type generates in every hour of 2013 to meet demand at the lowest total variable cost. Define your decision variables (with units), objective, and constraints: meeting demand each hour, each plant's hourly availability, ramping limits for the thermal types, and the hydro energy budget. Explain each constraint in a sentence.

Problem 1.2 (5 points)

Now add a carbon tax of τ dollars per tonne of CO_2. Show how it enters your objective. Then, **by hand**, find the tax at which coal falls behind each of the combined cycle, combustion turbine, and steam plants in the merit order.

Part 2: Dispatching New York in 2013 (20 points)

Problem 2.1 (10 points)

Solve your model in JuMP with no carbon tax. Report the total variable cost, annual CO_2 emissions, each plant type's share of generation, and any non-served energy. Compare your generation shares with New York's actual 2012 shares above, and give two reasons for the largest differences.

Problem 2.2 (10 points)

Plot hourly generation by plant type, stacked, for one week in January and one week in July. Explain the order in which plant types are used, and how the model uses hydro given its annual budget.

Part 3: Prices (15 points)

Problem 3.1 (8 points)

The shadow price of each hour's demand constraint is the cost of meeting one more MWh of demand in that hour: the marginal price of electricity. Compute it for every hour and plot the prices sorted from highest to lowest. Count a plant type as setting the price in the hours whose price, rounded to the cent, equals its variable cost. Which plant type sets the price most often, and in how many hours do combustion turbines or steam plants set it?

Problem 3.2 (7 points)

Report the shadow price of the hydro budget and interpret it. Suppose enough extra water could be diverted through the Niagara turbines to generate one more MWh over the year: what would that MWh be worth, and why that value? Be careful with the sign.

Part 4: A Carbon Tax (25 points)

Problem 4.1 (12 points)

Solve the dispatch problem with carbon taxes of $0, $20, $40, $60, and $100 per tonne. Report annual emissions and each fossil plant type's share of generation at each tax in a table. Between which taxes do emissions drop, and do the drops match your answer to 1.2? Why does coal keep running after it falls behind combined cycle plants?

Problem 4.2 (13 points)

Is there any tax that brings emissions below 40 million tonnes per year by changing only which existing plants run? Identify the lowest emissions redispatch can reach, and explain what sets that floor. What happens to the average of the year's hourly prices as the tax rises, and who pays it?

Part 5: Building New Capacity (25 points)

The state can now build six kinds of new capacity: combined cycle, combustion turbines, wind, solar, hydro, and combined cycle with carbon capture. New capacity behaves like the existing plants of the same type: the same variable cost, emissions, and availability, and for the gas plants the same ramp fraction. Two of the six need more detail:

- **Hydro.** The only sites are existing dams that have no turbines, so at most 295 MW can be built. We assume each MW of new hydro generates, on average, as much as each MW of existing hydro: each new MW raises the annual hydro generating capacity by 24,572,000 MWh ÷ 4,276 MW ≈ 5,746 MWh.
- **Combined cycle with carbon capture.** This is the fleet file's last row. It captures 90% of its CO_2, so it emits 0.05 t/MWh. It costs $39/MWh to run, more than a combined cycle plant, because it burns more gas and pays to move and store the CO_2 it captures.

Annualized costs of new capacity are in `data/new_capacity_costs.csv` (in $ per MW of capacity per year): `Cost2013` is from 2013 cost estimates and `CostToday` from 2025 ones.

```
new_costs = DataFrame(CSV.File("data/new_capacity_costs.csv"))
hydro_build_limit = 295   # most new hydro that can be built [MW]
```

New Capacity: Data and Cautions

The annualized costs are capital costs from the U.S. Energy Information Administration's 2013 and 2025 reports on new generating plants, spread over 30 years at 7%, plus fixed operating costs. The 295 MW of hydro is Oak Ridge National Laboratory's 2012 estimate of the capacity that New York's dams without turbines could support. Capture's variable cost of $39/MWh includes about $4.50/MWh to move and store the 0.45 t of CO_2 it captures from each MWh, at $10 per tonne.

Keep two things in mind when you interpret your results. First, the 2013 cost of capture is an estimate for a kind of plant that had not been built at commercial scale. Second, the two reports describe different hydro plants, 500 MW in 2013 and 100 MW in 2025, so part of the increase in hydro's cost between them comes from comparing a larger plant with a smaller one, not from a change in what the same plant costs.

Problem 5.1 (8 points)

Extend your model to include build decisions, using 2013 costs. Explain what changes in the objective and the constraints. With no emissions policy, what does the model build, and why? Compare the cost of a MWh from new wind with the cost of the fuel it would replace.

Problem 5.2 (8 points)

Cap annual emissions at 40 million tonnes, still with 2013 costs. What does the model build, what does the system cost, and what is the shadow price of the cap? Explain what the new plants replace.

Problem 5.3 (9 points)

Repeat 5.2's capped problem with today's costs. What gets built now, and what is the shadow price of the same cap? Then remove the cap and instead impose a carbon tax equal to that shadow price. Compare the two solutions, and explain the result using what you know about shadow prices. What does the change from 5.2 tell you about the role of a carbon price when technology costs fall?

Working With JuMP Output

`value.(x)` returns a JuMP container; use `value.(x).data` or index it with `value(x[g, t])` to get plain numbers for plotting. End cells that build large sets of variables or constraints with a semicolon to suppress long printed output. For a constraint `c`, `shadow_price(c)` reports how the objective changes as the constraint is relaxed; check its sign against your intuition. `fleet.RampFraction` is `missing` for the types with no ramp limit; test for that with `ismissing`.

References

List any external references consulted, including classmates.